SERRE'S AFFINENESS CRITERION AND COHOMOLOGY OF SEPARATED SCHEMES

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Abstract. These are the notes prepared by Yiming Wang for the talk *Serre's affineness criterion and cohomology of separated schemes* for the seminar on *Cohomology of sheaves and schemes* organized by Marc Hoyois during the summer semester 2026 at the University of Regensburg. These notes follow the perspective to algebraic geometry as developed in [Hoy26] and will freely use the notations and the results developed in this material.

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1. Functorial point of view

As mentioned in the introduction, we will adopt the approach to algebraic geometry as in [Hoy26]. We will shortly explain our perspective on sheaf cohomology. Recall first the following theorem from [Hoy26], which allows us to pass between the functor-of-points perspective on schemes and the classical perspective of locally ringed spaces.

The proofs of Proposition 1.1 and Proposition 1.2 were developed during the lectures corresponding to [Hoy26]. A proof-inclusive work-in-progress version of [Hoy26] is available from the author upon request.

Theorem 1.1 (Schemes as locally ringed spaces)**.** *Let* $\mathcal{P}\mathrm{rim}$ *be the collection of locally ringed spaces*

$$(\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)}),$$

where R *is any ring. Then the functor*

$$(|-|, \mathcal{O})\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{LRS}$$

restricts to an equivalence of categories

$$\mathrm{Sch} \xrightarrow{\sim} \mathrm{LRS}(\mathcal{P}\mathrm{rim}).$$

Under this equivalence, the following proposition states that the quasi-coherent modules correspond to the quasi-coherent sheaves of $\mathcal{O}_X$-modules.

Proposition 1.2 (Quasi-coherent modules as sheaves)**.** *Let* X *be a scheme. Then the functor*

$$\mathrm{Mod}_X \to \mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Shv}(|X|)), \quad M \mapsto (U \mapsto \Gamma(U, M))$$

is fully faithful and identifies Mod_X *with the category of quasi-coherent sheaves on* $(|X|, \mathcal{O}_X)$.

Sheaf cohomology and Čech cohomology are then constructed as a family of compatible functors from $\mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Shv}(|X|))$ to the category of abelian groups. For a quasi-coherent module M over a scheme X, we

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denote its corresponding quasi-coherent sheaf of $\mathcal{O}_X$-modules by $\widetilde{M}$. Then the sheaf cohomology and Čech cohomology of M are defined as the associated cohomologies of $\widetilde{M}$.

Remark 1.3 (A warning on derived global sections). In Proposition 1.2, the category Mod_X of quasi-coherent modules over X is indeed a Grothendieck abelian category having enough injectives (see [Sta26, Tag 077K]). Therefore, it makes sense to speak of the derived functor of the global section functor $\Gamma(X,-)\colon \mathrm{Mod}_X \to \mathrm{Ab}$. However, it is not a priori clear that this is the same as the derived functor of $\Gamma(X,-)\colon \mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Shv}(|X|)) \to \mathrm{Ab}$ restricted to Mod_X[1]. It is this latter derived functor that we adopt as the definition of sheaf cohomology.

2. Serre's affineness criterion

In this section, we prove Serre's affineness criterion. We begin by recalling the notion of an affine morphism and by proving a general lemma concerning affineness.

Definition 2.1 (Relative Spec and relative affineness)**.** Let X and Y be algebraic functors.

(a) Let A be a quasi-coherent algebra over X. The algebraic functor $\mathrm{Spec}(A)$ over X is defined by

$$\mathrm{Spec}(A)(R) = \{(x,a) \mid x \in X(R) \text{ and } a \text{ is a section of } \mathrm{Spec}(A(x)) \to \mathrm{Spec}(R)\}.$$

This defines a functor $\mathrm{Spec}\colon \mathrm{CAlg}_X^{\mathrm{op}} \to \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X}$.

(b) Let $f\colon Y \to X$ be a map of algebraic functors. We say that f is *affine*, or that Y is *affine* over X, if there exists $A \in \mathrm{CAlg}_X$ and an isomorphism $Y \simeq \mathrm{Spec}(A)$ over X. We denote by Aff_X the category of algebraic functors affine over X.

The following proposition gives a criterion to verify affineness of a map and gives a relative version of the algebro-geometry duality:

Proposition 2.2 (Characterization of affine morphisms)**.** *Let X be an algebraic functor.*

(a) A map $Y \to X$ is affine if and only if, for every ring R and every map $\mathrm{Spec}(R) \to X$, the pullback $Y \times_X \mathrm{Spec}(R)$ is affine.

(b) The functor Spec *defines an equivalence of categories*

$$\mathrm{Spec}\colon \mathrm{CAlg}_X^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_X,$$

whose inverse sends $f\colon Y \to X$ to $f_*(\mathcal{O}_Y)$.

In the lectures corresponding to [Hoy26], we remarked without proof that affineness is Zariski-local on the base. In the classical literature, this locality statement is often proved using the affineness criterion. Here, however, we use the locality of affineness to prove the affineness criterion. To avoid circularity, we include below an independent proof of this fact, using descent for quasi-coherent algebras.

Proposition 2.3. *Affineness of morphisms of schemes is Zariski-local on the base.*

Proof. Let $f\colon Y \to X$ be a map of schemes and $U \subseteq X$ an open subscheme. By Proposition 2.2(a) and the pasting law for pullback squares, if $Y \to X$ is affine, then $Y \times_X U \to U$ is again affine.

Conversely, suppose that $(U_i \subseteq X)_{i\in I}$ is an open covering and $Y \times_X U_i \to U_i$ is affine for each $i \in I$. By the Zariski descent for quasi-coherent algebras, the following diagram of categories is a limit diagram:

$$\mathrm{CAlg}_X \to \prod_{i\in I} \mathrm{CAlg}_{U_i} \rightrightarrows \prod_{i,j\in I} \mathrm{CAlg}_{U_i\times_X U_j} \substack{\rightrightarrows \\ \rightrightarrows} \prod_{i,j,k\in I} \mathrm{CAlg}_{U_i\times_X U_j\times_X U_k}.$$

Since the equivalence of categories in Proposition 2.2 is natural in X and the involution $(-)^{\mathrm{op}}\colon \mathrm{Cat} \to \mathrm{Cat}$ is an equivalence, the following diagram of categories is also a limit diagram:

$$\mathrm{Aff}_X \to \prod_{i\in I} \mathrm{Aff}_{U_i} \rightrightarrows \prod_{i,j\in I} \mathrm{Aff}_{U_i\times_X U_j} \substack{\rightrightarrows \\ \rightrightarrows} \prod_{i,j,k\in I} \mathrm{Aff}_{U_i\times_X U_j\times_X U_k}.$$

[1] In special cases, the derived functors coincide when restricted to Mod_X, e.g. when X is qcqs.

Since "affineness" is closed under base change, the family of affine maps $(Y \times_X U_i \to U_i)_{i\in I}$ can be glued into an affine map $\tilde{f}\colon \widetilde{Y} \to X$. Since $\coprod_{i\in I} U_i \to X$ is a Zariski-local epimorphism between schemes, it is an effective epimorphism in $\mathrm{Shv}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$. Therefore, $\mathrm{Shv}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})_{/X}$ is the limit of the following diagram[2]:

$$\prod_{i\in I} \mathrm{Shv}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})_{/U_i} \rightrightarrows \prod_{i,j\in I} \mathrm{Shv}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})_{/U_i\times_X U_j} \overset{\rightarrow}{\rightarrow} \prod_{i,j,k\in I} \mathrm{Shv}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})_{/U_i\times_X U_j\times_X U_k}.$$

By Zariski descent, we obtain an isomorphism $Y \simeq \widetilde{Y}$ over X, hence $Y \to X$ is affine. □

Lemma 2.4 (A criterion for affineness)**.** *Let X be a scheme. Suppose that there exist $f_1, \dots, f_n \in \mathcal{O}(X)$ such that $(f_1, \dots, f_n) = \mathcal{O}(X)$ and the open subfunctors $\mathrm{D}(f_i) \subseteq X$ are affine schemes. Then $X \simeq \mathrm{Spec}(\mathcal{O}(X))$.*

Proof. By the adjunction between $\mathcal{O}$ and Spec, the identity map $\mathrm{id}_{\mathcal{O}(X)}$ corresponds to a map

$$a\colon X \to \mathrm{Spec}(\mathcal{O}(X)).$$

Since the principal open subfunctors are given by taking pullbacks along $\mathbf{G}_m \hookrightarrow \mathbf{A}^1$, we find that $a^{-1}(\mathrm{D}(f_i)) = \mathrm{D}(f_i) \subseteq X$ for every $i = 1, \dots, n$. Recall that $(f_1, \dots, f_n) = \mathcal{O}(X)$ implies that the principal open subfunctors

$$(\mathrm{D}(f_i) \subseteq \mathrm{Spec}(\mathcal{O}(X)))_{i\in I}$$

form an open covering. Since base change of an open covering is again an open covering, $(\mathrm{D}(f_i) \subseteq X)$ is an affine open covering of X, where the affineness follows by assumption. By Proposition 2.2(a), the map

$$X \supseteq \mathrm{D}(f_i) \to \mathrm{D}(f_i) \subseteq \mathrm{Spec}(\mathcal{O}(X))$$

is affine, since both sides are affine schemes. By Proposition 2.3, we conclude that the map $a\colon X \to \mathrm{Spec}(\mathcal{O}(X))$ is affine. Being affine over an affine scheme, X is affine. Since X lies in the essential image of the right adjoint Spec, the unit map $a\colon X \to \mathrm{Spec}(\mathcal{O}(X))$ is an isomorphism. □

Lemma 2.5. *Let T be a nonempty quasi-compact sober topological space. Then T contains a closed point.*

Proof. Consider the following descending chain of nonempty closed subsets in T:

$$\dots \subseteq F_3 \subseteq F_2 \subseteq F_1 \subseteq F_0$$

For every $i_1 < i_2 \dots < i_n$, the finite intersection $\bigcap_{k=1}^n F_{i_k}$ must be F_{i_n}. Therefore, the family of closed subsets $\{F_0, F_1, \dots\}$ satisfies the finite intersection property. Since T is quasi-compact, the intersection of all of the closed subsets in the above chain is closed and nonempty. Therefore, the above chain stablizes and by Zorn's lemma, there exists a minimal closed subset E in T. We claim that E is irreducible. In fact, if $U, V \subseteq E$ are nonempty open subsets such that $U \cap V = \varnothing$, then $E \setminus U \subseteq E$ would be a closed subset of T, which contradicts the minimality of E. Let $x \in E$. Then $\overline{\{x\}} \subseteq E$ is a closed subset of T, hence $\overline{\{x\}} = E$ by the minimality of E. Therefore, every point $x \in E$ is a generic point of E. Since $E \subseteq T$ is an irreducible closed subset, there exists a unique generic point. In particular, E is a singleton. □

Theorem 2.6 (Serre's affineness criterion)**.** *Let X be a quasi-compact scheme. Then the following are equivalent:*

(a) X is affine;
(b) $\mathrm{H}^i(X, M) = 0$ for every $i > 0$ and $M \in \mathrm{Mod}_X$;
(c) $\mathrm{H}^1(X, I) = 0$ for every $I \in \mathrm{Idl}_X$.

Proof. (a) ⇒ (b): we have already seen the proof of this result, (see [Sta26, Tag 01XB]).

(b) ⇒ (c): when X is a scheme, I is a subobject of $\mathcal{O}_X$ in Mod_X. In particular I is a quasi-coherent module over X. The statement follows immediately.

[2]The result follows by embedding the sheaf categories into the corresponding ∞-categories of anima-valued sheaves, using the internal characterization of topoi, and observing that 0-truncatedness of morphisms can be detected after pullback along an effective epimorphism.

(c) $\Rightarrow$ (a): we will first show that every closed point has an affine principal neighbourhood. Let $p \in |X|$ be a closed point. Being a scheme, X is covered by affine open subfunctors, hence we may choose an affine open neighbourhood $U \subseteq X$ of p. Since $|-|$ is an order preserving bijection between closed reduced subschemes and closed subsets of the topological space $|X|$, there exists a reduced closed subscheme $Y = X \setminus U$ of X, whose underlying closed subset is $|X| \setminus |U| \subseteq |X|$. Since $\mathrm{V}(-)$ is an order reversing bijection between quasi-coherent ideals and closed subschemes, there exists a quasi-coherent ideal $I_Y \in \mathrm{Mod}_X$ such that $I_Y \subseteq \mathcal{O}_X$ and $\mathrm{V}(I_Y) = Y$. Furthermore, let $Y \cup \{p\}$ be the reduced closed subscheme of X whose underlying closed subset of $|X|$ is $|Y| \cup \{p\}$ and let $I_{Y\cup\{p\}}$ be its corresponding quasi-coherent ideal. Since $|Y| \subseteq |Y| \cup \{p\}$, there is an inclusion $I_{Y\cup\{p\}} \subseteq I_Y$. Since Mod_X is abelian, there exists an exact sequence

$$0 \to I_{Y\cup\{p\}} \to I_Y \to I_Y/I_{Y\cup\{p\}} \to 0.$$

Recall that for an open immersion $j\colon U \hookrightarrow X$, the restriction functor $j^*\colon \mathrm{Mod}_X \to \mathrm{Mod}_U$ is exact, and restricts to quasi-coherent ideals, i.e., the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Idl}_X & \hookrightarrow & \mathrm{Mod}_X \\ {\scriptstyle j^*}\downarrow & & \downarrow{\scriptstyle j^*} \\ \mathrm{Idl}_U & \hookrightarrow & \mathrm{Mod}_U\,. \end{array}$$

Furthermore, it is straightforward to see that $j^*\colon \mathrm{Idl}_X \to \mathrm{Idl}_U$ behaves well with respect to correspondence between quasi-coherent ideals and closed subschemes. Therefore, by applying j^*, the following sequence is exact:

$$0 \to I_p \to \mathcal{O}_U \to \mathcal{O}_U/I_p \to 0 \quad \text{in } \mathrm{Mod}_U,$$

since $Y \cap U = \varnothing$ and $(Y \cup \{p\}) \cap U = \{p\}$. Since U is affine, applying the global section functor $\Gamma(U, -)\colon \mathrm{Mod}_U \simeq \mathrm{Mod}_{\mathcal{O}(U)}$ gives the following exact sequence in $\mathrm{Mod}_{\mathcal{O}(U)}$:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U) \to \kappa(p) \to 0.$$

Let $V = |X| \setminus \{p\}$. Observe that $\{U, V\}$ is an open covering of X. By the sheaf condition for quasi-coherent sheaves, the following diagram of abelian groups is a pullback square:

$$\begin{array}{ccc} \Gamma(X, I_Y/I_{Y\cup\{p\}}) & \longrightarrow & \Gamma\left(U, \mathcal{O}_U/I_p\right) \\ \downarrow & & \downarrow \\ \Gamma\left(V, (I_Y/I_{Y\cup\{p\}})|_V\right) & \longrightarrow & \Gamma\left(U\cap V, (I_Y/I_{Y\cup\{p\}})|_{U\cap V}\right). \end{array}$$

Since $Y \cap V = Y = (Y \cup \{p\}) \cap V$, the lower terms of the above diagram are zero, hence the lower horizontal map is an isomorphism and the upper horizontal map is again an isomorphism. In other words, $\Gamma(X, I_Y/I_{Y\cup\{p\}}) \simeq \kappa(p)$. By the long exact sequence of sheaf cohomology, the following sequence is exact:

$$\Gamma(X, I_Y) \to \kappa(p) \to \mathrm{H}^1(X, I_{Y\cup\{p\}}),$$

where the rightmost term vanishes by assumption. Therefore, the map $\Gamma(X, I_Y) \to \kappa(p)$ is surjective, and there exists $f \in \mathcal{O}(X)$ such that the image of f is invertible in $\kappa(p)$. Since $\mathrm{D}(f) \subseteq X$ can be written as $\mathrm{D}(f) \simeq X \times_{\mathbf{A}^1} \mathbf{G}_m$, we find that $p \in \mathrm{D}(f)$. By pulling back along every $x\colon \mathrm{Spec}(R) \to X$, it follows that $\mathrm{D}(f) \subseteq \mathrm{D}(I_Y)$. Since $\mathrm{D}(I_Y)$ is the open complement of Y, it follows from the inclusion below that $\mathrm{D}(f) \subseteq U$, hence affine

$$|X| \setminus |U| = |Y| = |\mathrm{V}(I_Y)| = |X| \setminus |\mathrm{D}(I_Y)| \subseteq |X| \setminus |\mathrm{D}(f)|.$$

We will show that the principal affine open subfunctors of X form an open covering of X. Denote W the union of all $|\mathrm{D}(f)|$ such that $\mathrm{D}(f) \subseteq X$ is affine. It suffices to show that $|X| = W$. Suppose that $W \neq |X|$, then $T = |X| \setminus W$ is a nonempty closed subset of X. Since X is quasi-compact, $|X|$ is a quasi-compact topological space, and the closed subspace T is again quasi-compact. Being a closed subspace of a sober space, T is again sober. Therefore, by Lemma 2.5, T contains a closed point. However, by what we've shown before, W contains

all closed points. By contradiction, we conclude that $|X| = W$. Since X is quasi-compact, there exists a finite open covering $(\mathrm{D}(f_i) \subseteq X)_{i=1,\dots,r}$ consisting of principal affine open subfunctors.

By Lemma 2.4, it suffices to show that $(f_1, \dots, f_r) = \mathcal{O}(X)$. We first claim that the map $(f_1, \dots, f_r) \colon \mathcal{O}_X^{\oplus r} \to \mathcal{O}_X$ is an epimorphism in Mod_X. Since epimorphisms in Mod_X can be checked after restriction to an affine open cover, it suffices, for each inclusion $j_i \colon \mathrm{D}(f_i) \hookrightarrow X$, to show that the map

$$j_i^*(f_1, \dots, f_r) \colon \mathcal{O}_{\mathrm{D}(f_i)}^{\oplus r} \to \mathcal{O}_{\mathrm{D}(f_i)}$$

is an epimorphism in $\mathrm{Mod}_{\mathrm{D}(f_i)}$. After applying the global section $\Gamma(\mathrm{D}(f_i), -)$, it suffices to show that the map

$$j_i^*(f_1, \dots, f_r) \colon \mathcal{O}(\mathrm{D}(f_i))^{\oplus r} \to \mathcal{O}(\mathrm{D}(f_i)), \quad (a_1, \dots, a_r) \mapsto \sum_{j=1}^{r} a_j f_j|_{\mathrm{D}(f_i)}$$

is an epimorphism, i.e., surjective, in $\mathrm{Mod}_{\mathcal{O}(\mathrm{D}(f_i))}$. This follows from the fact that $f_i|_{\mathrm{D}(f_i)} \colon \mathrm{D}(f_i) \to \mathbf{A}^1$ factors through $\mathbf{G}_m$, hence is invertible. Consider the following short exact sequence in the abelian category Mod_X:

$$0 \to K \to \mathcal{O}_X^{\oplus r} \xrightarrow{(f_1,\dots,f_r)} \mathcal{O}_X \to 0,$$

where $K = \ker(f_1, \dots, f_r)$. Define K_i via the following pullback square:

$$\begin{array}{ccc} K_i & \longrightarrow & \mathcal{O}_X^{\oplus i} \\ \downarrow & & \downarrow \\ K & \longrightarrow & \mathcal{O}_X^{\oplus r}, \end{array}$$

where $\mathcal{O}_X^{\oplus i}$ denotes the first i components of $\mathcal{O}_X^{\oplus r}$. By the pasting law for pullback squares, the following commutative squares are cartesian:

$$\begin{array}{ccccc} K_{i-1} & \hookrightarrow & K_i & \hookrightarrow & K \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{O}_X^{\oplus(i-1)} & \hookrightarrow & \mathcal{O}_X^{\oplus i} & \hookrightarrow & \mathcal{O}_X^{\oplus r}. \end{array}$$

In particular, the following give us a filtration of K in Mod_X:

$$0 = K_0 \hookrightarrow K_1 \hookrightarrow K_2 \hookrightarrow K_3 \hookrightarrow \dots \hookrightarrow K_{r-1} \hookrightarrow K_r = K.$$

Consider the projection to the i-th factor $\mathrm{pr}_i \colon \mathcal{O}_X^{\oplus i} \to \mathcal{O}_X$. By pasting law again, the following commutative squares are cartesian:

$$\begin{array}{ccccc} K_{i-1} & \hookrightarrow & \mathcal{O}_X^{\oplus(i-1)} & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow \\ K_i & \hookrightarrow & \mathcal{O}_X^{\oplus i} & \xrightarrow{\mathrm{pr}_i} & \mathcal{O}_X. \end{array}$$

Therefore, the canonical map $K_i/K_{i-1} \hookrightarrow \mathcal{O}_X$ is a monomorphism. Since quasi-coherent ideals over X corresponds to subobjects of $\mathcal{O}_X$ in Mod_X, K_i/K_{i-1} is a quasi-coherent ideal over X. By the short exact sequence

$$0 \to K_{i-1} \to K_i \to K_i/K_{i-1} \to 0,$$

we obtain the corresponding long exact sequence of cohomology groups

$$\dots \to \mathrm{H}^1(X, K_{i-1}) \to \mathrm{H}^1(X, K_i) \to \mathrm{H}^1(X, K_i/K_{i-1}) \to \dots.$$

The right term vanishes by assumption. Therefore, the middle term vanishes if the left term vanishes. By induction, we conclude that $\mathrm{H}^1(X, K) = 0$. Consider exact sequence induced by the short exact sequence before:

$$\mathcal{O}(X)^{\oplus r} \xrightarrow{(f_1,\dots,f_r)} \mathcal{O}(X) \to \mathrm{H}^1(X, K) = 0.$$

Therefore, the left map in $\mathrm{Mod}_{\mathcal{O}(X)}$ is surjective and $(f_1, \dots, f_r) = \mathcal{O}(X)$. □

3. Cohomology of separated schemes

In the last section, we will prove that the Čech cohomology of an affine open covering agrees with the sheaf cohomology when the scheme is separated. The key ingredients we use is that every finite intersection of affine open subfunctors is again affine, hence has trivial higher Čech cohomology.

Lemma 3.1. *Let X be a separated scheme and $U, V \subseteq X$ be affine open subschemes of X. Then $U \cap V$ is affine.*

Proof. By definition, the map $\Delta_X : X \to X \times X$ is a closed immersion. Consider the following pullback square:

$$\begin{array}{ccc} U \cap V & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle \Delta_X} \\ U \times V & \longrightarrow & X \times X. \end{array}$$

Since closed immersions are closed under base change, the map $U \cap V \hookrightarrow U \times V$ is a closed immersion. Since affine schemes are closed under limits, $U \times V$ is affine, hence $U \cap V$ is affine. □

Theorem 3.2. *Let X be a separated scheme and $\mathcal{U} = (U_i \subseteq X)_{i \in I}$ be an affine open covering. For every $i \geq 0$, there is a natural isomorphism $\check{\mathrm{H}}^i(\mathcal{U}, M) \xrightarrow{\sim} \mathrm{H}^i(X, M)$ functorial in $M \in \mathrm{Mod}_X$.*

Proof. For every $n \geq 0$, we have already constructed the natural comparison maps

$$\check{\mathrm{H}}^n(\mathcal{U}, M) \to \mathrm{H}^n(X, M)$$

and shown that they are isomorphisms for $n = 0, 1$. Suppose that the result holds for each $n < k$. Recall that M can be embedded in a flasque, quasi-coherent module N. Consider the following short exact sequence:

$$0 \to M \to N \to P \to 0.$$

For every finite intersection $U_{i_0,\dots,i_p}$, the restriction functor is exact and the following sequence is exact:

$$0 \to M|_{U_{i_0,\dots,i_p}} \to N|_{U_{i_0,\dots,i_p}} \to P|_{U_{i_0,\dots,i_p}} \to 0.$$

By the long exact sequence for sheaf cohomology, the following sequence is exact:

$$0 \to \Gamma(U_{i_0,\dots,i_p}, M|_{U_{i_0,\dots,i_p}}) \to \Gamma(U_{i_0,\dots,i_p}, N|_{U_{i_0,\dots,i_p}}) \to \Gamma(U_{i_0,\dots,i_p}, P|_{U_{i_0,\dots,i_p}}) \to \mathrm{H}^1(U_{i_0,\dots,i_p}, M|_{U_{i_0,\dots,i_p}}).$$

By Lemma 3.1, Theorem 2.6 and induction, the finite intersection $U_{i_0,\dots,i_p}$ is affine and the term $\mathrm{H}^1(U_{i_0,\dots,i_p}, M|_{U_{i_0,\dots,i_p}})$ vanishes. Therefore, the following sequence of Čech complexes is exact:

$$0 \to \check{\mathrm{C}}^*(\mathcal{U}, M) \to \check{\mathrm{C}}^*(\mathcal{U}, N) \to \check{\mathrm{C}}^*(\mathcal{U}, P) \to 0.$$

Since the comparison maps are natural, there exists an induced map between long exact sequences:

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & \check{\mathrm{H}}^{k-1}(\mathcal{U}, N) & \longrightarrow & \check{\mathrm{H}}^{k-1}(\mathcal{U}, P) & \longrightarrow & \check{\mathrm{H}}^k(\mathcal{U}, M) & \longrightarrow & \check{\mathrm{H}}^k(\mathcal{U}, N) & \longrightarrow & \dots \\ & & \simeq\downarrow & & \simeq\downarrow & & \downarrow & & \downarrow & & \\ \dots & \longrightarrow & \mathrm{H}^{k-1}(X, N) & \longrightarrow & \mathrm{H}^{k-1}(X, P) & \longrightarrow & \mathrm{H}^k(X, M) & \longrightarrow & \mathrm{H}^k(X, N) & \longrightarrow & \dots, \end{array}$$

where the two vertical maps are isomorphisms as displayed, by the induction hypothesis. Recall that flasque sheaves are acyclic, and the i-Čech cohomology vanishes for $i > 0$. Therefore, the above diagram becomes

$$\begin{array}{ccccccc} 0 & \longrightarrow & \check{\mathrm{H}}^{k-1}(\mathcal{U}, P) & \xrightarrow{\sim} & \check{\mathrm{H}}^k(\mathcal{U}, M) & \longrightarrow & 0 \\ & & \simeq\downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^{k-1}(X, P) & \xrightarrow{\sim} & \mathrm{H}^k(X, M) & \longrightarrow & 0, \end{array}$$

where the horizontal maps are isomorphisms, by exactness. Therefore, the natural map

$$\check{\mathrm{H}}^k(\mathcal{U}, M) \to \mathrm{H}^k(X, M)$$

is an isomorphism. The statement then follows by induction. □

Corollary 3.3. *Let X be a separated scheme. For every $i \geq 0$, there exist natural isomorphisms*

$$\check{\mathrm{H}}^i(X, M) \to \mathrm{H}^i(X, M)$$

functorial in $M \in \mathrm{Mod}_X$.

Proof. Let $(U_i \subseteq X)_{i \in I}$ be an open covering of X. For every $U_i \subseteq X$, there exists an affine open covering $(V_{ij} \subseteq U_i)_{j \in J_i}$. These assemble into an affine open covering $(V_{ij} \subseteq X)_{i \in I, j \in J_i}$. In other words, every open covering of X can be refined to an affine open covering of X, i.e., the poset of affine open coverings is cofinal in the poset of all open coverings. Recall that the canonical map in the statement is induced by the filtered colimit:

$$\operatorname*{colim}_{\mathcal{U} \in \mathrm{Cov}(X)} \check{\mathrm{H}}^i(\mathcal{U}, M) \to \mathrm{H}^i(X, M).$$

By Theorem 3.2 and the above, the canonical map above is an isomorphism. □

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